

StyleCap

Grounded Multi-Style Video Captioning

Containerized Track 2 agent: watches short clips, extracts visual facts, writes four distinct caption styles - each grounded in what is actually on screen.

Gemma 3 12B

AMD MI300X

ROCm + vLLM

Fireworks AI

Docker GHCR

GRADED IMAGE

ghcr.io/quantumbyte-01/amd-track2-captioner:v34

GITHUB

github.com/QuantumByte-01/amd-track2-captioner

CONTACT

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The Problem

- > Caption each 30s-2min clip in 4 styles: formal, sarcastic, humorous-tech, humorous-non-tech
- > Docker container, ~12 clips, 10-minute wall clock, valid JSON schema, no runtime API key
- > Generic captions that could fit any video score near 0.6 accuracy - specificity wins
- > Any model allowed for Track 2; leaderboard scored by automated LLM judge

FORMAL

Objective, detail-rich sentences grounded in visible facts

SARCASTIC

Dry irony - Ah yes... cadence from reference winners

HUMOROUS-TECH

One dev metaphor anchored to a real visual detail

HUMOROUS-NON-TECH

Relatable When you finally... line, still clip-specific

Built for the Grader

Accuracy + style match on every caption

FACTUAL ACCURACY

Vision fact sheet before styling. Captions trace back to frames.

STYLE REGISTER

Prompts calibrated to reference cadence and per-style length caps.

RELIABILITY

Degradation ladder + schema repair. Never empty or invalid JSON.

SPEED

6 parallel workers, batched judge, 90s/clip budget, 9-min soft deadline.

PARTNER PRIZE (\$3K GEMMA)

Same pipeline end-to-end on google/gemma-3-12b-it via vLLM on AMD MI300X. Demo video + GEMMA_RESULTS.md in repo.

Fact Sheet First

Two-phase anti-hallucination pipeline

StyleCap Pipeline Architecture

Two-phase anti-hallucination - parallel per-clip workers



HOW IT WORKS

Per-Clip Pipeline

1 Download

Parallel fetch across 6 workers

2 Sample

ffmpeg uniform-seeking, 6 frames at 768px

3 Ground

Vision model produces strict JSON fact sheet

4 Write

GLM batch: 2 candidates per style

5 Judge

One batched call picks best per style

6 Repair

Length policy + schema validation + ladder

6

frames / clip

2

candidates

~90s

clip budget

540s

soft deadline

86s

Gemma 8-clip

8/8

eval strong

Clip v2

Orange kitten among green foliage in a garden, looking at camera

Formal

A young orange tabby kitten sits among dense green foliage in an outdoor setting, looking directly at the camera with an alert and curious expression.

Sarcastic

A kitten outdoors, clearly plotting something elaborate and fully confident it will succeed.

Humorous Tech

A small autonomous agent has entered the garden environment and is scanning for input. Next action: unknown. Rollback plan: none.

Humorous Non Tech

A tiny cat has gone outside and is now judging everything it sees with great authority.

SAMPLE OUTPUT

Clip v7

Person chopping vegetables (zucchini) in a kitchen

Formal

A person is chopping zucchini into small cubes on a wooden cutting board.

Sarcastic

Ah yes, the ancient art of chopping zucchini... truly the pinnacle of culinary skills.

Humorous Tech

When you try to refactor your code but end up with too many slices instead of clean functions.

Humorous Non Tech

When you realize you actually have to chop the veggies before dinner can happen.

Gemma on AMD MI300X

Identical pipeline - end-to-end on the pod

- > google/gemma-3-12b-it via vLLM + ROCm on Instinct MI300X (~82% VRAM)
- > Ground, write, judge all on Gemma - no Fireworks deploy (\$7/hr avoided)
- > 8 public clips in 86 seconds - GEMMA_RESULTS.md in repo
- > Demo: rocm-smi + vLLM logs + live caption generation

GEMMA OUTPUT (v1 autumn boulevard)

Formal

Urban street with traffic, buses, and yellow-leaf trees beside gray apartment towers.

Sarcastic

Ah yes, a perfectly organized urban commute where everything moves in the same direction.

Humorous Tech

When you refactor your codebase and everything still runs on the same lanes of traffic.

Humorous Non Tech

When you think you mastered parallel parking, then you see the traffic.

Graded Docker Image

Serverless models for post-deadline grader availability

Vision / Ground

`accounts/fireworks/models/kimi-k2p6`
Multimodal fact extraction from 6 frames

Caption / Judge

`accounts/fireworks/models/glm-5p2`
Style batch + batched candidate pick

Fallback

`glm-5p2-fast router`
JSON recovery under time pressure

Container

`linux/amd64, public GHCR`
`FIREWORKS_API_KEY` baked at build time

SUBMIT THIS IMAGE ON LABLAB

`ghcr.io/quantumbyte-01/amd-track2-captioner:v34`

Why StyleCap Wins

- + Accuracy-first: every caption traces to a vision JSON fact sheet
- + Style-calibrated: reference-winning cadence baked into write + judge prompts
- + Reliable: degradation ladder always returns schema-valid results.json
- + Fast: parallel workers + batched judge fits the 10-minute container limit
- + AMD story: Gemma-3-12B proven on MI300X with recorded demo evidence

Repository + submission

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Thank you